Kinetic Investigations in High-Pressure Region on Selected Catalysts for Ammonia Reforming

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The environmental impact of excessive fossil fuel use is recognized as one of the most pressing global challenges. Transitioning to green energy requires efficient energy carriers to replace fossil fuels. A common approach is to store surplus renewable energy as hydrogen via electrolysis. However, the challenge lies in finding an efficient method for hydrogen transport. The ideal intermediate substance must fulfill three key criteria: efficient conversion of hydrogen to the intermediate, cost-effective large-scale transport, and efficient hydrogen release to meet fluctuating energy demands. Among potential candidates, ammonia (NH_3) stands out as a promising intermediate due to the well-established Haber-Bosch process, its high gravimetric and volumetric hydrogen capacity, and its ease of liquefaction (9 bar at 25°C) [1]. In this context, the BMBF-funded AmmoRef project, under which this research is conducted, aims to advance energy-efficient, CO_2-free NH_3 reforming through process optimization and catalyst development [2].

To attain high purity and pressurized H_2 from initially liquid NH_3, the catalytical reforming process is carried out in the gas-phase. Conventionally, ruthenium-based catalysts deliver the highest performance and the reaction is run at ambient pressures due volume expansion. H_2 is subsequently separated and compressed to target gas specifications, reducing process efficiency. [1] Another approach is to reform at elevated pressures, which requires special reactor and catalyst materials since high temperatures are required to reach full conversion due to endothermic reaction. High-pressure reforming is yet an uncharted region, with works [4-7] being the only reports, where also only Ru-based catalysts are used at pressures up to 40 bar and temperatures up to 550 °C. Kinetic studies, using both macro and microkinetic approaches, are predominantly conducted at atmospheric pressure [8], highlighting the need for validation of industrially applicable kinetic models at high-pressure conditions.

In the frame of this, various catalysts are tested in a pressure and temperature range of 1 to 50 bar and 500 – 750 °C respectively. Feed composition is varied between 95 % NH_3 and H_2 co-feeding to increase the precision of determined reaction orders. The data is fitted using Temkin-Pyzhev and modified power-law type rate equations. Sensitivity of both models to

increased pressure are investigated. Furthermore, the required model complexity to predict the outcome with a single set of kinetic parameters on a wide p, T and $y_{i,in}$ range is evaluated. It is shown that when a modified power-law equation with a thermodynamic equilibrium correction term is used, a very good fit is obtained in 1 – 15 bar and 500 – 650 °C range for both Ni / Al_2O_3 and Fe_3Mo_3N catalysts. In higher pressure and temperature, a deviation from these kinetic parameters is observed, where Temkin-Pyzhev formalism is not necessarily sustained. The findings reported in this work, are to be used in further techno-economical analysis of small scale (50 Nm^3_{H2} h^{-1}) high-pressure NH3 reforming plant in various modes of operation.

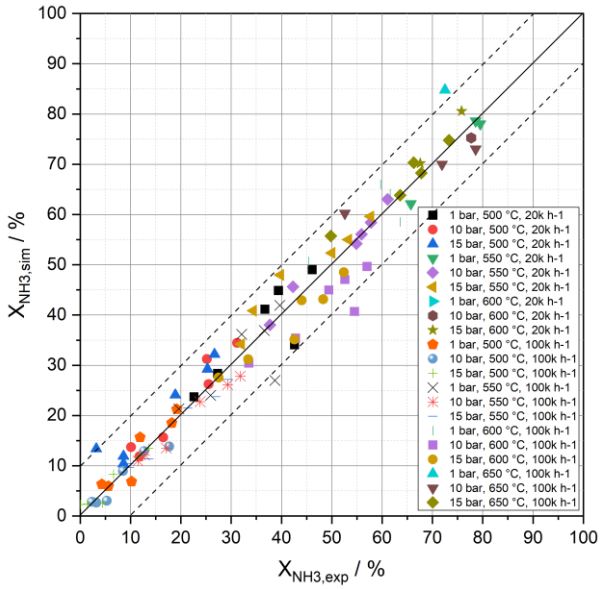

Figure 1: Parity plot of the Ni / Al_2O_3 catalyst using a modified power-law type reaction rate equation.

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